CSYE 7374 · LLMOps · Status update

NPS Guide Copilot

Huiling Fu · November 4, 2026 · week of the Neon login and saved trips

Project summary

NPS Guide Copilot is a U.S. national-park trip planner. The website plans the trip with its own data and code. A local Python Copilot understands the request, asks follow-up questions, and explains the plan. The model does not replace the planner.

Goal this week: keep a person's trip and Copilot conversation after they leave the browser, and give each person their own login.

Recent update:

- Neon Postgres is connected from the Next.js site with DATABASE_URL.
- Neon Auth handles signup and login. The account page is /account. Users are stored by Neon in the neon_auth schema.
- A logged-in trip is saved in user_trips. A Copilot turn is saved in chat_threads. Without a login, the trip stays in this browser only.
- Checked this afternoon: one account, one session, and one saved trip are in the database. A chat row appears after the next logged-in Copilot turn.

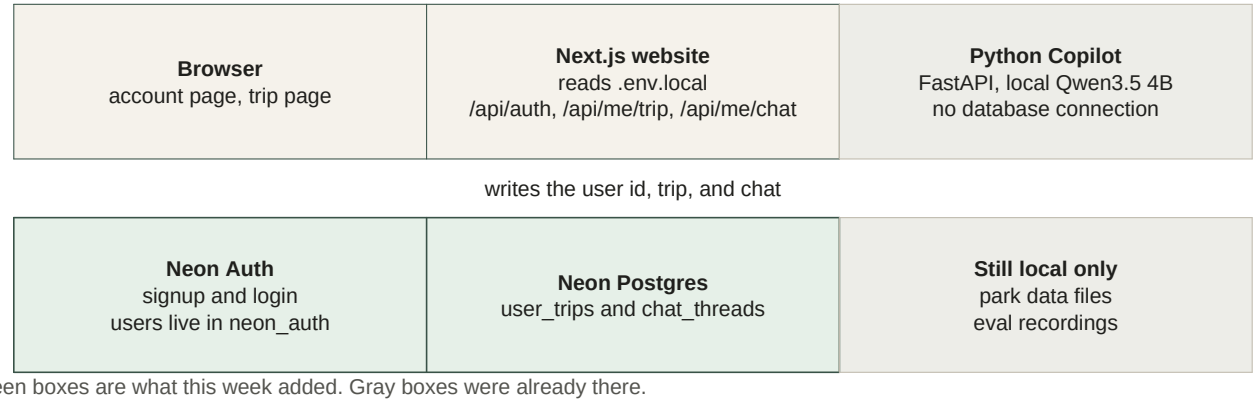
Team members

Name	Role	This week
Jun Zhai	Repository owner. Copilot pipeline, eval gates, and the planner.	No change.
Tianyue Chen	Teammate (JadenChen10).	No change.
Huiling Fu	Website data, then Neon login and saved trips.	This update.

No one joined or left. Formal role split is still the one already in the project notes.

Architecture

The new pieces are the two Neon boxes. The Python Copilot is unchanged and does not open a database connection. Next.js saves the reply after Copilot returns it.



Green boxes are what this week added. Gray boxes were already there.

Tech stack

- Website: Next.js 16, deployed on Vercel. Trip planning stays in TypeScript.
- Copilot: Python FastAPI and a local Qwen3.5 4B. CI replays recorded answers and does not call Neon.
- New: Neon Postgres and Neon Auth (managed Better Auth), through @neondatabase/serverless and @neondatabase/auth.
- Copied once into .env.local: DATABASE_URL and NEON_AUTH_BASE_URL. The cookie secret is already generated. The JWKS URL is not used.

Technical demo

About four minutes. Show the local site, then the same rows in the Neon console.

- Open the account page and sign up with an email and an 8-character password. Neon Auth rejects anything shorter.
- In the Neon console, show the new user under Better Auth.
- Open the trip page while logged in and show the user_trips row. Logging out leaves the on-screen planner usable. The saved copy stays in Neon.
- With the local Copilot running, send one message and show the new chat_threads row.

Plan for next week

- Add DATABASE_URL, NEON_AUTH_BASE_URL, and the cookie secret to the Vercel project.
- Allow the deployed site address in Neon Auth, then sign in there and reload the saved trip.
- Finish one logged-in Copilot conversation so chat_threads has a real thread, not only the trip row.

Blockers / risks

- The Copilot process is still local. The Vercel site can save a trip, and it can save a chat only when that process is reachable.
- Neon Auth currently trusts the local site address. The public Vercel address is not allowed until next week's step.
- Password length is fixed by Neon Auth at 8 characters. The app cannot set a shorter rule.

One-minute script

This week I connected Neon to the website, not to the Python Copilot. I copied two values into .env.local: the database connection string and this project's Auth URL. The site now has signup and login. After login, the trip is stored in Neon, and each Copilot reply is stored there too. I can already see the account and one saved trip in the Neon console. Next week I will put the same two values on Vercel so the deployed site can log in as well. The Copilot model still runs on my machine, so chat history on the public site waits until that service is reachable.